Fiscal Year:	FY 2007	**Task Last Updated:**	FY 05/31/2008
PI Name:	Ezekoye, Ofodike Ph.D.		
Project Title:	Resonant Acoustic Control of Turbulent Diffusion Flames for Pollutant Reduction		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	None		
Space Biology Element:	None		
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Zip Code:	78712-1063	**Congressional District:**	12
Comments:			
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	A technology roadmap developed in 1999 by the industrial combustion community under funding from the Department of Energy identified three core R&D needs for industrial combustion over the next 20 years. The highest priority needs were identified to be burner technologies, boiler systems and process heating systems. These technologies impact a wide range of industrial applications, such as power generation, heat treatment, kilns, drying, copper smelting and glass melting. Burners represent a cross-cutting technology and improvements in burner technology will have major payoffs. The industrial combustion roadmap identifies constraints and requirements for next generation burners that include emissions constraints, stabilization issues, and economical premixing mechanisms. The roadmap further states that the advanced burner of the future “would also be able to change the flame pattern or heat delivery from the burner” and that “one concept of a ‘smart’ burner incorporates non-steady-state combustion where fluctuations are controlled rather than		

Task Description:	random." There is a clear need to scientifically explore all innovative burner concepts since it is not obvious what directions in burner designs will prove fruitful in the future. We are investigating an essentially unexploited burner strategy that uses high-intensity, high-frequency forcing of a fuel jet to achieve substantial partial-premixing of initially nonpremixed reactants. Our results show that it is possible to produce dramatic changes in the structure and luminosity of a turbulent nonpremixed jet flame with the application of strong forcing of the jet flow. For example, a turbulent nonpremixed methane jet flame, which exhibits substantial yellow luminosity due to emission from soot, can be made to turn completely blue and to shorten to about 1/3 of its original length, by the application of the forcing. These intense pulsations are achieved by modulating the fuel flow with a high-speed valve at a frequency that coincides with the organ-pipe resonance frequency of the fuel tube. The combination of rapid fuel flow rate modulation, coupled with the organ pipe resonance of the fuel tube results in very large amplitude, high –frequency modulation of the fuel gas pressure and velocity. The resulting flow exhibits a high degree of partial premixing of the fuel and air. Using direct flame photography, hotwire velocity measurements, planar laser induced fluorescence, and planar laser mie scattering we find that the changes in flame properties are due to ingestion (sucking) of ambient air into the fuel tube, vortex-ring entrainment, and turbulence generated by the unsteadiness. This form of high- amplitude, high-frequency forcing of the fuel jet differs significantly from most other forcing schemes, which tend to use low-amplitude, high-frequency (e.g. with a speaker) or high-amplitude, low-frequency (e.g. flame puffs) forcing. In this project we will study and detail the mechanisms responsible for significant partial premixing in resonantly driven, transitional and turbulent jet diffusion flames. The ability to predict the flame length, and lifted flame length for these pulsed flames, and also the ability to discuss transitions between the luminous and nonluminous flame conditions requires careful experimental work. These experiments are to be divided between low gravity and normal gravity conditions. Differences in flame properties between the low gravity conditions and normal gravity conditions will help us to decouple buoyancy-induced modifications to mixing and flame length from acoustically-induced modifications. Some normal gravity experiments will use conventional measurements of gas species simply to establish the basic characteristics of this type of flame. Numerical modeling will be used to better understand the experimental results. The objectives of this research are to: 1) characterize the mechanisms for increased near field mixing in strongly pulsed high frequency jet flames, 2) use the microgravity environment to develop an understanding of the role of buoyancy for near field mixed flames in modifying flame characteristics,
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research has the potential to substantially reduce pollutants formed in jet diffusion flames which benefits our national goals towards a cleaner environment.
Task Progress:	Control of nonpremixed flames for tunable radiative and emissions properties represents an ambitious goal for combustion science. One mechanism to control flame properties is the use of fuel flow pulsing. In order to generate sufficiently strong pulsations over a wide frequency range, it has been useful to use the organ pipe resonance of the fuel tube to drive the fuel flow oscillations. Over the past three years of NASA funding we have experimentally and computationally investigated the effects of buoyancy on the pulsed flow control of nonpremixed flames. We found that radiative emissions associated with soot production is strongly tied to residence time in either soot growth promoting or soot growth inhibiting zones of the flame. The level of forcing impacts the relative amount of air (and oxygen) mixed with the fuel flow, and this ultimately affects the environment that the soot particles evolve in. A difference between normal gravity and reduced gravity flames pulsed at a similar level is that the reduced gravity flames appear to be broader and more luminous at the flame base. The higher luminosity of low-gravity flames has been observed previously in laminar flames and is typically considered to be due to increased soot formation resulting from the lower strain rates associated with the absence of buoyant acceleration. We should emphasize, however, that it has been established that this resonantly-forced jet exhibits significant partial premixing of non-fuel gas into the fuel stream, and it is believed that the premixing plays a significant role in the reduction in the soot luminosity of the forced (as compared to unforced) conditions. Therefore, it is possible that the increase in the soot luminosity could have to do with the modification of the partial premixing mixing in the absence of buoyancy. For example, in low gravity it is likely that the combustion products in the region near the tube exit are not convected away. In this case, the reverse flow may not be as efficient at bringing oxygen into the fuel tube, but instead entrains combustion products. Summary and Objectives In this project we studied and detailed the mechanisms responsible for significant partial premixing in resonantly driven, transitional and turbulent jet diffusion flames. The ability to predict the flame length, and lifted flame length for these pulsed flames, and also the ability to discuss transitions between the luminous and nonluminous flame conditions has required the combination of careful experimental and computational work. Experiments were divided between low gravity and normal gravity conditions. Differences in flame properties between the low gravity conditions and normal gravity conditions helped us to decouple buoyancy-induced modifications to mixing and flame length from pulsed-flow-induced modifications. Some normal gravity experiments used conventional measurements of gas species simply to establish the basic characteristics of this type of flame. Numerical modeling was used to better understand the experimental results. The objectives of this research were to: 1) characterize the mechanisms for increased near field mixing in strongly pulsed high frequency jet flames, 2) use the microgravity environment to develop an understanding of the role of buoyancy for near field mixed flames in modifying flame characteristics, Summary of Findings Experimental Work We designed a system to include our resonant acoustic burner into our existing drop rig. The development time for incorporating the burner into the rig was somewhat longer than we anticipated because of space restrictions in the drop rig. Dozens of drops were performed for the resonantly-forced jet flames at three different mean-flow Reynolds

numbers (850, 1250, and 1850). In each case, the jet was fully-modulated at the organ-pipe resonant frequency of the tube. As discussed earlier, the strongly forced flames are blue and hence are not highly luminous; therefore, the imaging had to be performed at relatively low framing rates (500 Hz) and low shutter speeds. For both cases of normal and low gravity, as the Reynolds number is increased the flame appears shorter but wider. At the lower Reynolds number conditions, the structure of the flames is similar across gravity level over the lower half of the flame; however, the low-gravity flame becomes significantly more luminous farther downstream. The higher luminosity of low-gravity flames has been observed previously in laminar flames and is typically considered to be due to increased soot formation resulting from the lower strain rates associated with the absence of buoyant acceleration. We should emphasize, however, that it has been established that this resonantly-forced jet exhibits significant partial premixing and it is believed that the premixing plays some role in the reduction in the soot luminosity of the forced (as compared to unforced) conditions. Therefore, it is possible that the increase in the soot luminosity could have to do with the modification of the partial premixing mixing in the absence of buoyancy. For example, in low gravity it is possible that the combustion products in the region near the tube exit are not convected away. In this case, the reverse flow may not be as efficient at bringing oxygen into the fuel tube, but instead entrains combustion products. The clarification of this issue was a major goal of this research project.

Computational Work

We applied a finite volume CFDC (Computational Fluid Dynamics with Chemistry) code – UNICORN to study and simulate the strongly forced CH4–air diffusion flames. UNICORN (UNsteady Ignition and COmbustion with ReactioNs) solves the equations for the conservation of mass, momentum, individual species and energy in the unsteady formulation. The cylindrical form of the conservation equations are solved on a 2D axisymmetric staggered grid.

When going through the complete forcing cycle, the simulations clearly reveal that the forcing action serves to premix the fuel and air. The vortices generated by the forcing action bring in the cold air from the surroundings into the fuel stream, and thus premix the fuel and air. The reaction zone or the flame forms a cylinder, which surrounds the fuel-air mixture. The presence of oxygen in the internal part of the flame serves to oxidize the soot particles and thus result in dramatic soot reduction.

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